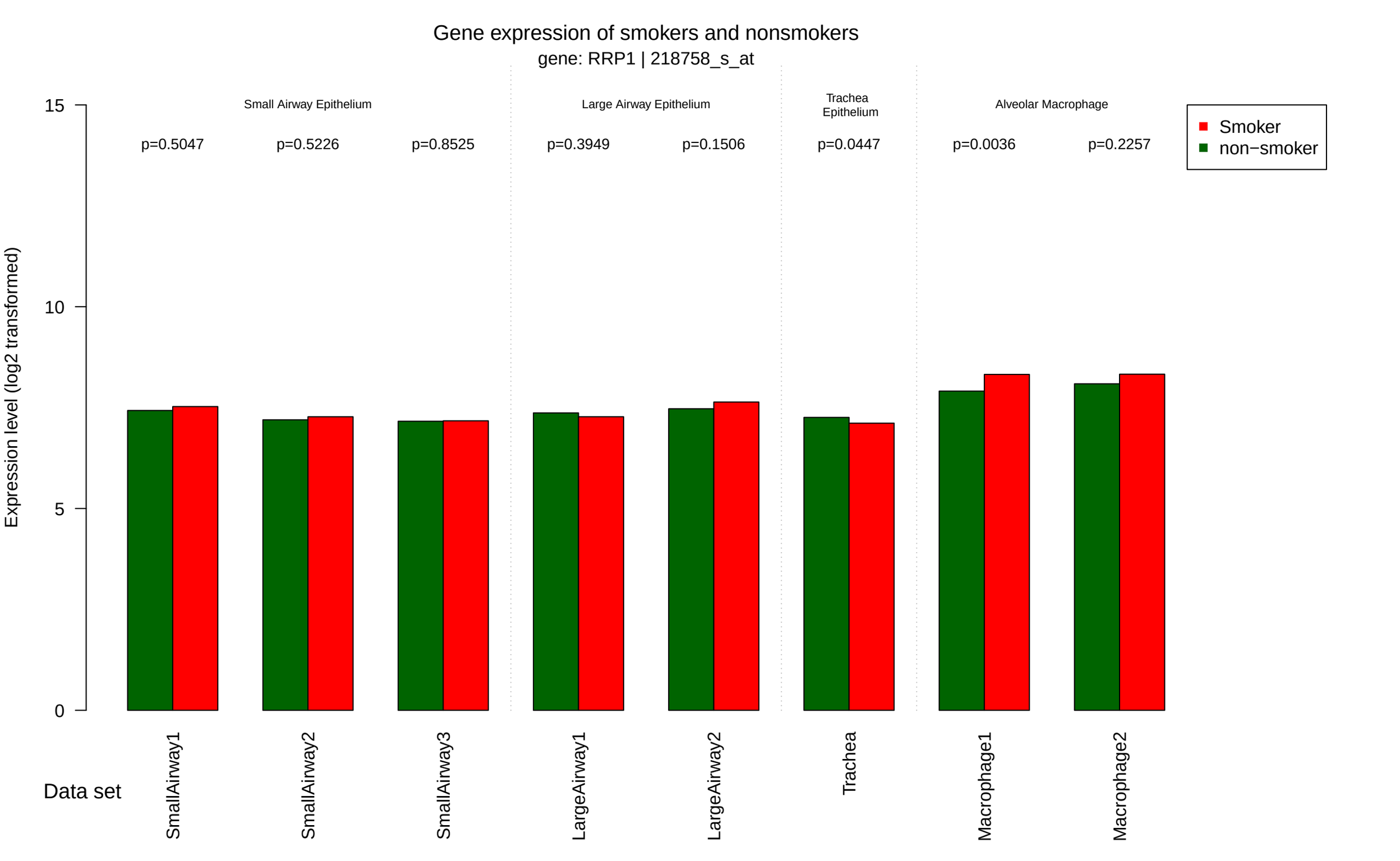
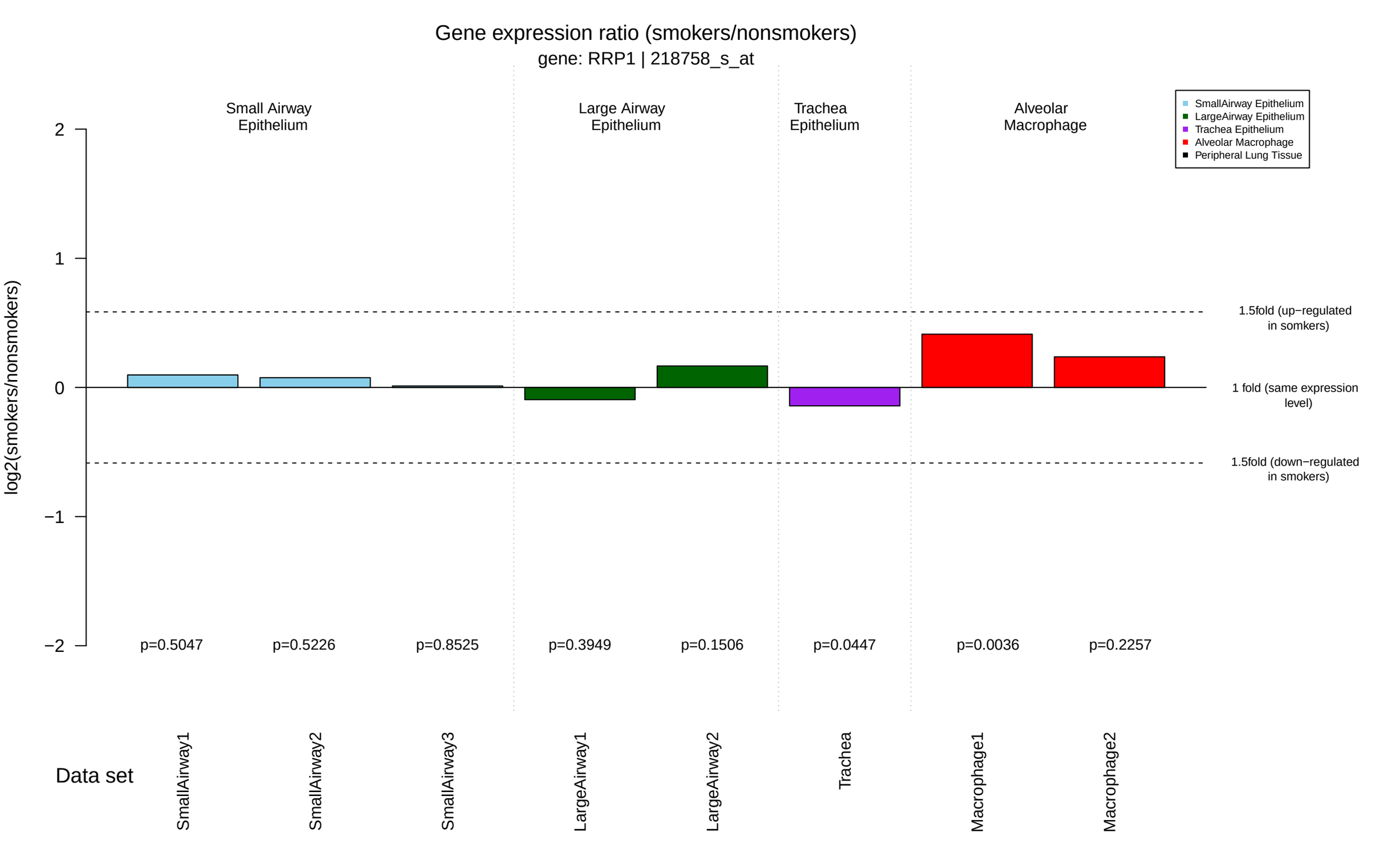
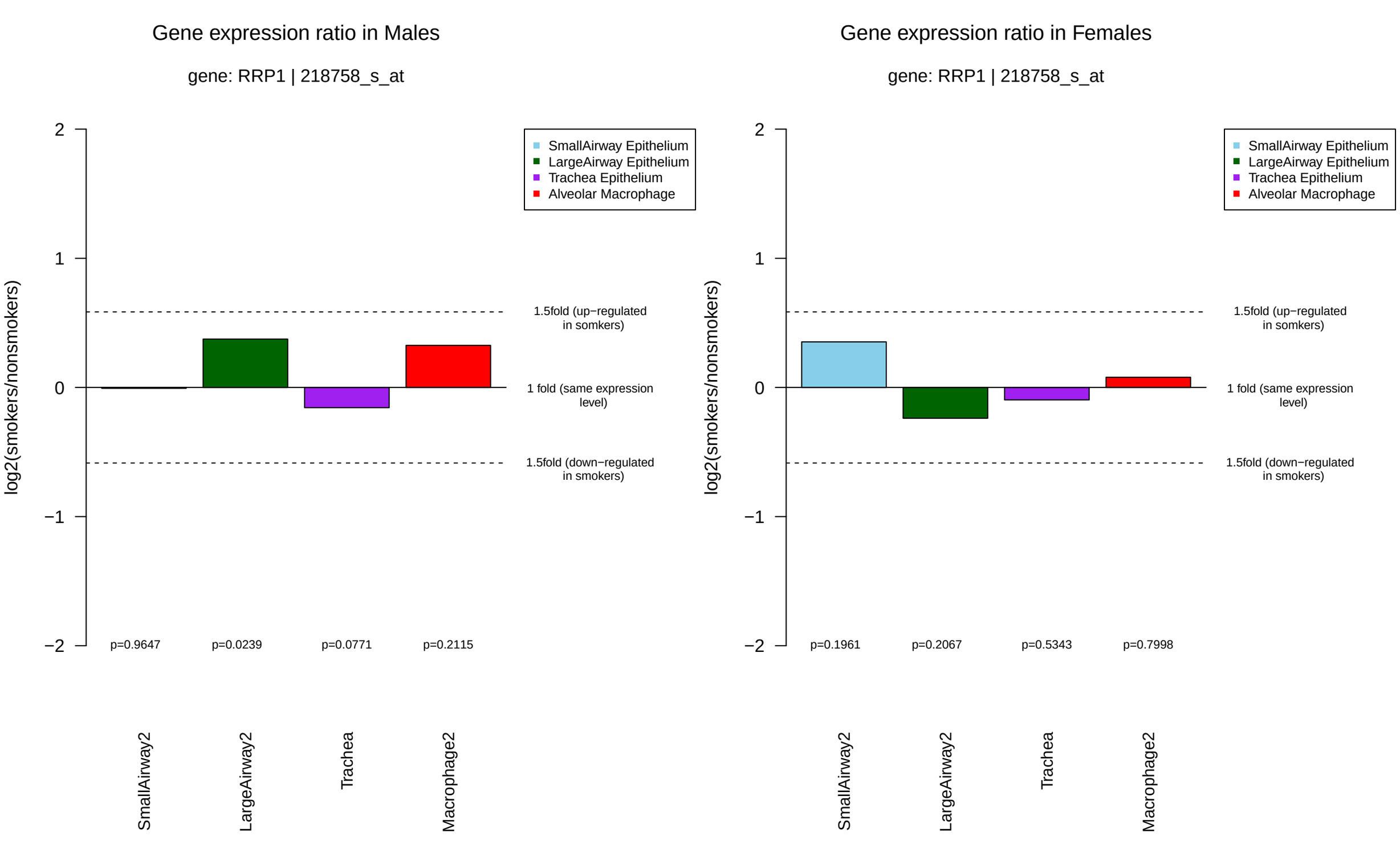
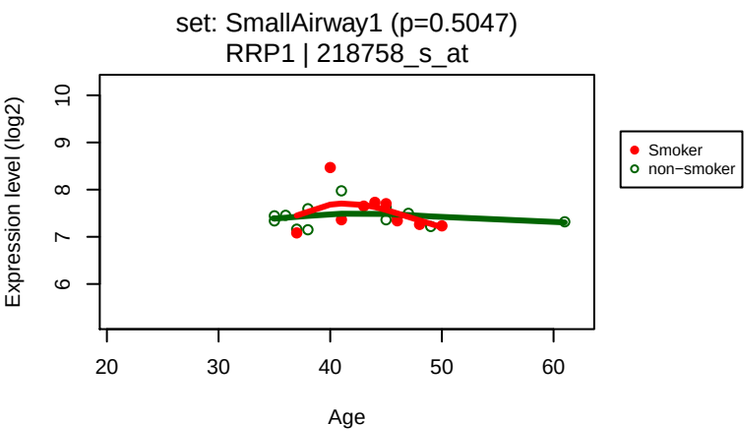
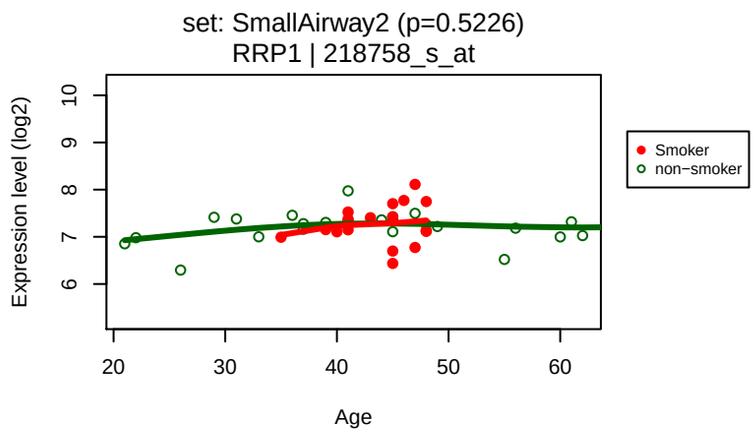
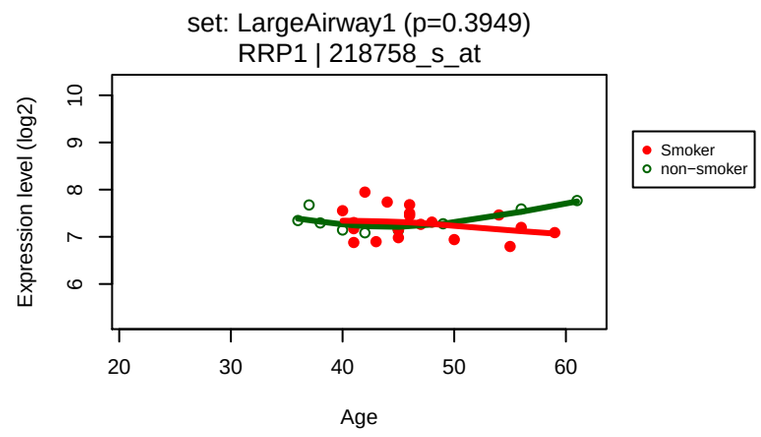

Gene expression of smokers and nonsmokers
gene: RRP1 | 218758_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5047
p=0.5226
p=0.8525
p=0.3949
p=0.1506
p=0.0447
p=0.0036
p=0.2257
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

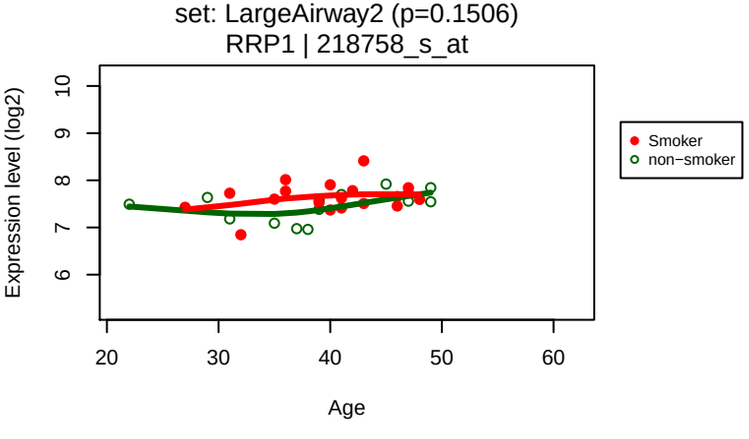

Gene expression ratio (smokers/nonsmokers)
gene: RRP1 | 218758_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5047
p=0.5226
p=0.8525
p=0.3949
p=0.1506
p=0.0447
p=0.0036
p=0.2257
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

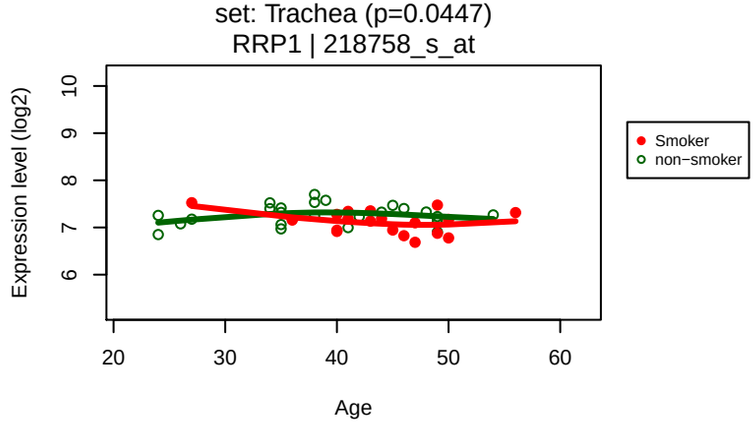

Gene expression ratio in Males
gene: RRP1 | 218758_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9647
p=0.0239
p=0.0771
p=0.2115
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRP1 | 218758_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1961
p=0.2067
p=0.5343
p=0.7998
SmallAirway2
LargeAirway2
Trachea
Macrophage2

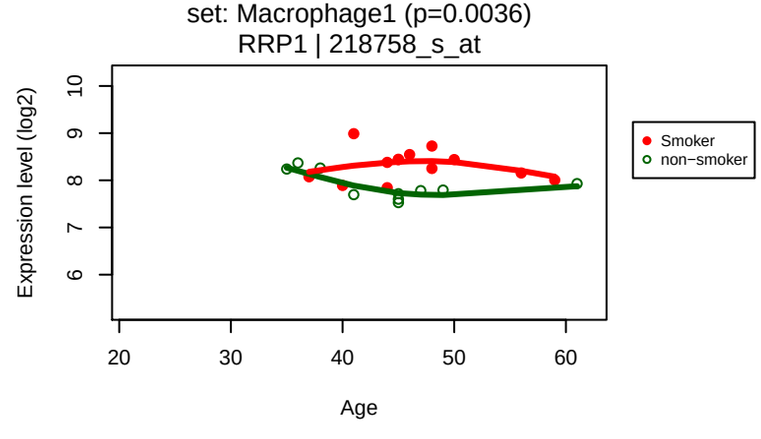

set: SmallAirway1 (p=0.5047)
RRP1 | 218758_s_at
Expression level (log2)
Age
Smoker
non−smoker

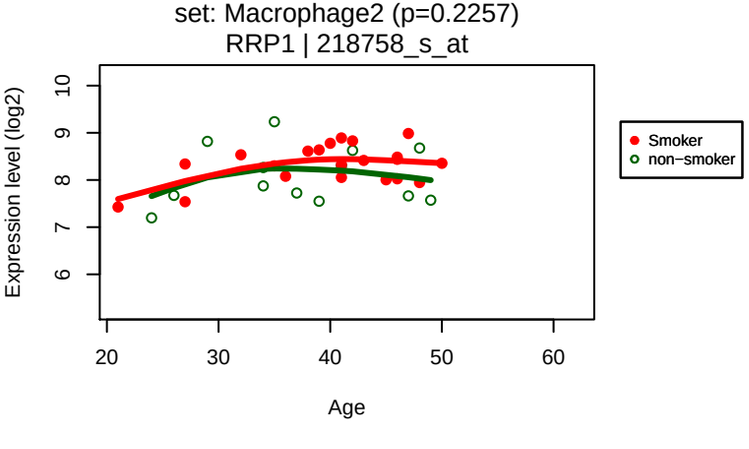

set: SmallAirway2 (p=0.5226)
RRP1 | 218758_s_at
Expression level (log2)
Age
Smoker
non−smoker

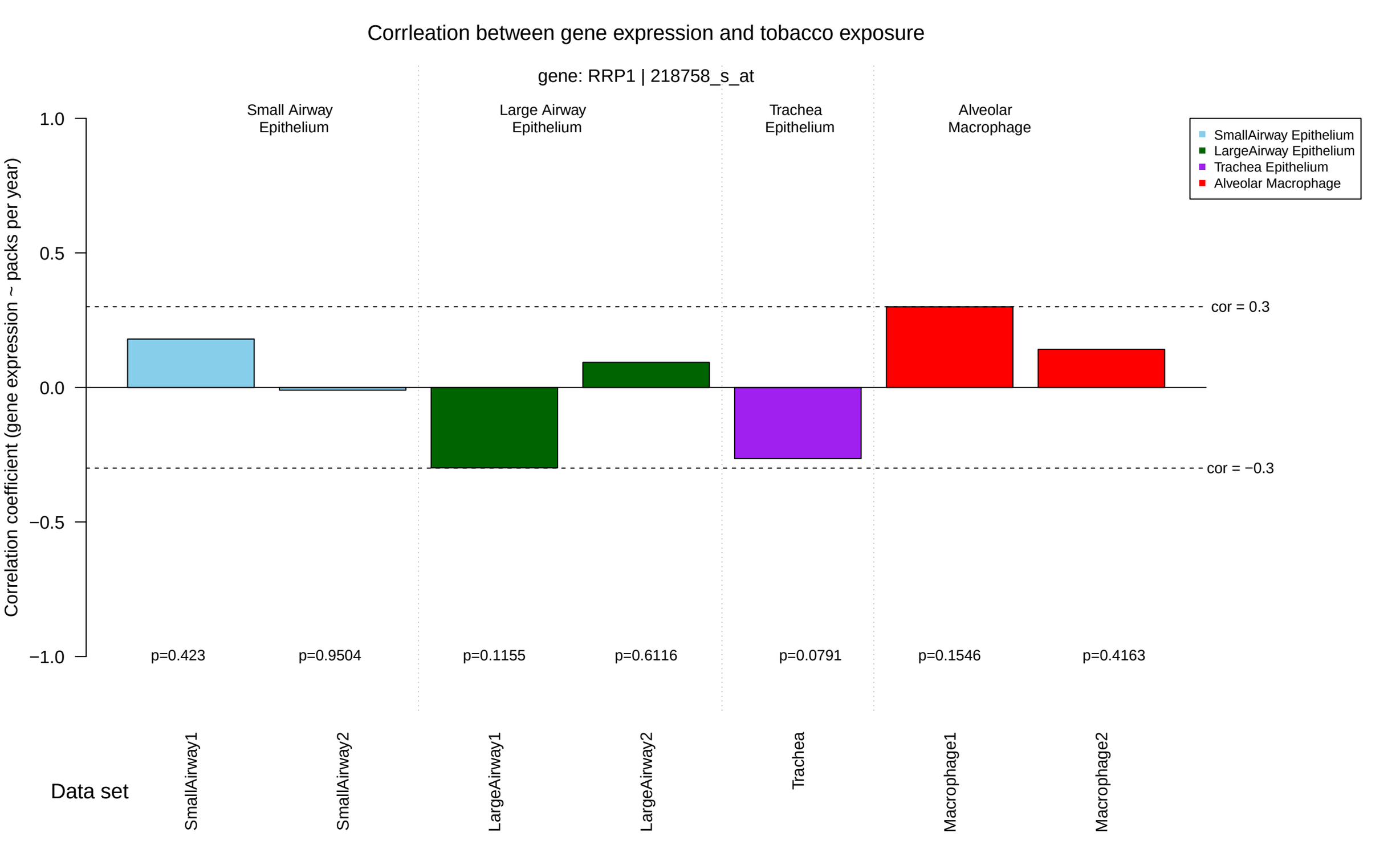

set: LargeAirway1 (p=0.3949)
RRP1 | 218758_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1506)
RRP1 | 218758_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0447)
RRP1 | 218758_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0036)
RRP1 | 218758_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2257)
RRP1 | 218758_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRP1 | 218758_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.423
p=0.9504
p=0.1155
p=0.6116
p=0.0791
p=0.1546
p=0.4163
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2